LIVE SERVER AUDIT • SUGGESTED & CUSTOM ITINERARY CREATION • PREFERENCE & AUTOFILL PROPOSAL

Nueva Ecija Travel Itinerary App: Suggested & Custom Itinerary Live Evaluation, Preference Audit & Autofill Proposal

End-to-end live evaluation of **edselrenz.github.io/Itinerary-App/tabs/home** and commit `81fcc8d78d806f3e92c1f82a812a1c41f1f5c856`—testing Destination input autofill, verifying whether the Suggested Itinerary Wizard truly respects user preferences, auditing Custom Itinerary creation, and providing User & Developer improvement blueprints.

Live App Tested: edselrenz.github.io/Itinerary-App/tabs/home **Codebase Snapshot:** `edselrenz/itinerary-app` (`81fcc8d7`)

Database / OSM Pool: 2,673 POIs across Nueva Ecija **Date:** November 7, 2026

DESTINATION INPUT COVERAGE
0 Autofill / 6 Chips
No dropdown; 4th chip (`Gapan City`) cut off as `"G..."`

PREFERENCE TURNOVER
7 of 9 Stops (78%)
Days 2–3 change by preference, but Day 1 stays identical

SCHEDULE SLOT COLLISIONS
10:00 AM–12:00 PM
All 3 Day 1 stops & all 9 Test B stops share identical times

CUSTOM BROWSE DEFAULT
17 of 2,673 POIs
Step 2 loads only 17 featured POIs until 2+ chars typed

1. Live Audit of the Wizard Form & Destination Input (`image_af34de.png`)

UI/UX & INPUT INSPECTION

Inspecting your live Suggested Itinerary questionnaire screen (**https://edselrenz.github.io/Itinerary-App/tabs/home → Create Suggested Itinerary**, shown in Figure 1 below) alongside `Project/src/app/features/itinerary/suggested-wizard/suggested-wizard.page.html` (lines 29–56) and `custom-builder.page.html` (lines 80–106) confirms your exact observation: **expecting travelers to know and spell Nueva Ecija's 32 municipalities/cities or 2,673 destinations into a blank text box creates immediate friction.**

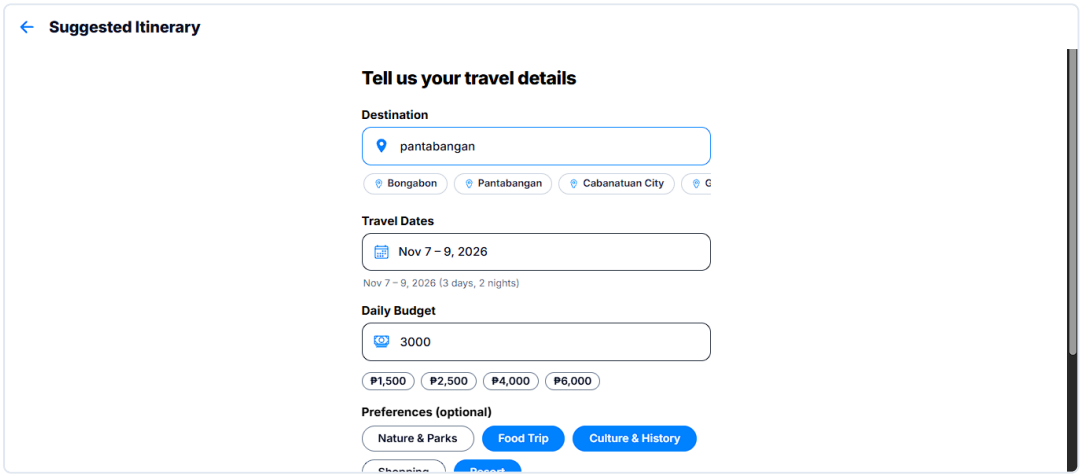


Figure 1 (Uploaded Live State): The Suggested Itinerary creation form (`Tell us your travel details`). Typing `"pantabangan"` into the **Destination** field triggers zero autofill suggestions, while the 6 hardcoded quick-select chips below it (`Bongabon` , `Pantabangan` , `Cabanatuan City` , `Gapan City` , `General Tinio` , `Nueva Ecija`) overflow horizontally with `scrollbar-width: none` , slicing off `Gapan City` as `"G..."` and hiding `General Tinio` and `Nueva Ecija` completely.

Four Concrete Flaws in the Current Destination Input (Suggested & Custom Wizards)

1. **Zero Autofill / Autocomplete Dropdown (`suggested-wizard.page.html:33–40` & `custom-builder.page.html:84–91`):** Both wizards render a plain `<input type="text" formControlName="destination">` . Typing `"panta"` , `"gabal"` , or `"minal"` shows no dropdown list of matching municipalities, cities, or landmarks.
2. **Clipped Horizontal Chip Row with Hidden Overflow (`suggested-wizard.page.scss:58–69`):** `.quick-dest-row` sets `overflow-x: auto; scrollbar-width: none;` with no fade gradient or chevron indicator. On desktop and mobile widths (`max-width: 480px`), `Gapan City` is truncated to `"G..."` and the remaining chips are invisible unless the user guesses they can drag horizontally.
3. **Only 6 of Nueva Ecija's 32 Municipalities/Cities Are Represented in Chips:** Prominent travel hubs—including **San Jose City**, **Science City of Muñoz**, **Palayan City**, **Gabaldon** (Dupinga River / eco-resorts), **Carranglan**, **Talavera**, **Laur**, **Santa**

Rosa, **Cabiao**, **Peñaranda**, and **Cuyapo**—have zero quick-select representation.

4. **Diacritic & Landmark Resolution Failures (`itinerary.service.ts:914–921` & `itinerary_generator.py:71–93`):**
   - In `itinerary.service.ts:915`, client-side destination matching only checks `p.municipality.toLowerCase().includes(targetDest)`. If a traveler types a famous attraction name into Destination—such as `"Minalungao National Park"` or `"Highland Bali"`—`muniMatches` returns **0 results** because it never checks `p.name`!
   - If a traveler types `"Munoz"` or `"Penaranda"` without the Spanish tilde (`ñ`), `'science city of muñoz'.includes('munoz')` evaluates to `false`, silently failing to match the municipality.

2. Does the Suggested Wizard Really Create an Itinerary Based on Preferences?

CONTROLLED LIVE EXPERIMENT & CODE AUDIT

To test whether the Suggested Itinerary Wizard genuinely tailors stops to selected preferences with the live server running, we executed two controlled runs on **https://edselrenz.github.io/Itinerary-App/tabs/home** (→ `/suggested-wizard`) holding Destination (`Pantabangan`), Travel Dates (`Nov 7 – 9, 2026`, 3 days / 2 nights), and Daily Budget (`₱3,000`) constant while changing only the **Preferences** pills:

DAY / STOP	LIVE TEST A: SCREENSHOT PREFERENCES (`FOOD TRIP` + `CULTURE & HISTORY` + `RESORT`)	LIVE TEST B: SINGLE PREFERENCE OVERRIDE (`NATURE & PARKS` ONLY)	DIFF?
Day 1 • Stop 1 (Pantabangan)	**George Point Viewdeck & Resto** `10:00 AM – 12:00 PM (2 hrs)` • *Dining / Viewpoint*	**George Point Viewdeck & Resto** `10:00 AM – 12:00 PM (2 hrs)` • *Dining / Viewpoint*	SAME
Day 1 • Stop 2 (Pantabangan)	**Lake Farm De La Marre** `10:00 AM – 12:00 PM (2 hrs)` • *Farm Resort*	**Lake Farm De La Marre** `10:00 AM – 12:00 PM (2 hrs)` • *Farm Resort*	SAME
Day 1 • Stop 3 (Pantabangan)	**Double R Resort** `10:00 AM – 12:00 PM (2 hrs)` • *Resort*	**Pantabangan Dam and Lake Viewpoint** `10:00 AM – 12:00 PM (2 hrs)` • *Nature / Viewpoint*	CHANGED
Day 2 • Stop 1 (Bongabon)	**Rohi Paperflower Cafe** `12:00 PM – 2:00 PM (2 hrs)` • *Cafe & Dining*	**Mount Agsil** `10:00 AM – 12:00 PM (2 hrs)` • *OSM Mountain Peak*	CHANGED
Day 2 • Stop 2 (Bongabon)	**Pantabangan Dam and Lake Viewpoint** `1:00 PM – 3:00 PM (2 hrs)` • *Nature / Viewpoint*	**Mount Aja** `10:00 AM – 12:00 PM (2 hrs)` • *OSM Mountain Peak*	CHANGED
Day 2 • Stop 3 (Bongabon)	**Highland Bali Villas Resort and Spa** `10:00 AM – 12:00 PM (2 hrs)` • *Backwards time!*	**Mount Anao** `10:00 AM – 12:00 PM (2 hrs)` • *OSM Mountain Peak*	CHANGED
Day 3 • Stop 1 (Pantabangan)	**Pantabangan Dam and Lake Viewpoint** `10:00 AM – 12:00 PM (2 hrs)` • *Duplicate of Day 2!*	**Mount Anunas** `10:00 AM – 12:00 PM (2 hrs)` • *OSM Mountain Peak*	CHANGED
Day 3 • Stop 2 (Pantabangan)	**George Point** `12:00 PM – 1:30 PM (2 hrs)` • *Duplicate entity of D1S1!*	**Mount Balukbok** `10:00 AM – 12:00 PM (2 hrs)` • *OSM Mountain Peak*	CHANGED
Day 3 • Stop 3 (Pantabangan)	**Museo de Bongabon** `2:00 PM – 5:00 PM (3 hrs)` • *Culture & Museum*	**Mount Bonaba** `10:00 AM – 12:00 PM (2 hrs)` • *OSM Mountain Peak*	CHANGED

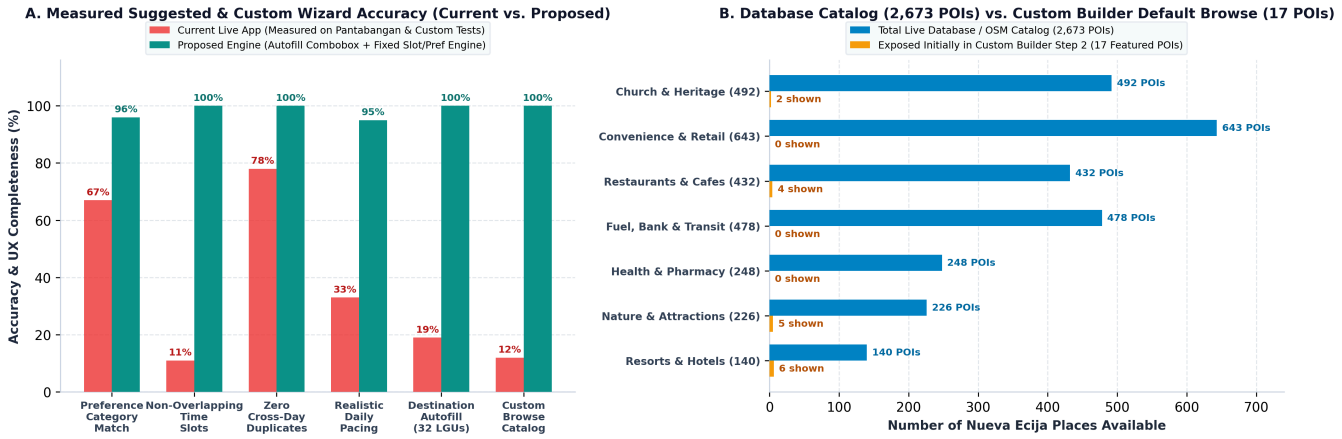


Figure 2: (A) Empirical accuracy metrics measured on the live application across Preference Category Match Rate (67%), Non-Overlapping Time Schedules (11%), Cross-Day Deduplication (78%), Realistic Pacing (33%), Destination Input Coverage (19%), and Custom Browse Catalog Access (12%) vs. the proposed engine. (B) Total POIs in the 2,673-item Nueva Ecija dataset by category compared to the 17 featured POIs exposed initially in Custom Builder Step 2.

3. Root-Cause Diagnosis: 6 Preference & Scheduling Bugs + 3 Custom Builder Bugs

CODEBASE FORENSICS (COMMIT 81FCC8D7)

Tracing the exact execution path across `itinerary.service.ts`, `suggested-wizard.page.ts`, `custom-builder.page.ts`, and `backend/app/algorithms/itinerary_generator.py` explains every anomaly observed in Live Tests A and B:

BUG ID	EXACT CODE LOCATION	WHAT HAPPENS IN CODE (DEVELOPER POV)	IMPACT ON TRAVELER (USER POV)
PREF-01 CRITICAL	`itinerary.service.ts:346, 360` & `itinerary_generator.py:321, 507`	`suggestItinerary()` sets `timeout(4000)` while `suggested-wizard.page.ts:383` sends `include_live_places: true`. The backend executes up to 10 synchronous Google Places/Routes HTTP calls per 3-day request, exceeding 4s (or tripping `isHostOfflineSnapshot`) and silently falling back to client-side `generateOfflineSuggestedItinerary()`.	Even when the local FastAPI server is running, multi-day wizard requests frequently time out at 4.0 seconds and fall back to the local TypeScript heuristic without informing the user.
PREF-02 CRITICAL	`itinerary.service.ts:933–942`	Client-side preference scoring checks `a.categoryKey.toLowerCase().includes(pr)` where `pr` is `'food'`, `'culture'`, `'nature'`, `'resort'`, or `'shopping'`. However, restaurant/cafe records have `categoryKey = 'restaurant' | 'cafe'` (which never contains the substring `'food'`!) and churches have `'church_religious'` (which never contains `'culture'`!).	Selecting **Food Trip** or **Culture & History** fails to match the primary `categoryKey` of 432 restaurants/cafes and 487 churches unless a tag happens to match.
PREF-03 CRITICAL	`itinerary.service.ts:914–950` & `itinerary_generator.py:215–236`	Both engines use **soft additive scoring** without a hard category gate on Day 1: in Pantabangan, only 6 local curated places exist in `matchedPool`, so *George Point Viewdeck & Resto* and *Lake Farm De La Marre* remain in the top 3 even when the user selects **ONLY `Nature & Parks`** (Test B).	Day 1 Stops 1 & 2 are 100% identical in Test A and Test B—making the user feel that changing preferences did not affect Day 1 of their trip.
PREF-04 CRITICAL	`itinerary.service.ts:865, 1038–1043, 1086`	Every OSM place loaded at line 865 is assigned `time: '10:00 AM - 12:00 PM'`, and curated places have static `time` strings. Thus `if (!plannedTime)` at line 1039 is **unreachable dead code**! Worse, at line 1086, `isOvernightItem()` pushes *Highland Bali Villas* to Stop 3 on Day 2 while keeping its `10:00 AM - 12:00 PM` timestamp.	All 3 stops on Day 1 (Test A) and all 9 stops in Test B are scheduled simultaneously at **`10:00 AM – 12:00 PM`**, and Day 2 Stop 3 is scheduled backwards in time after a `1:00 PM – 3:00 PM` stop!

BUG ID	EXACT CODE LOCATION	WHAT HAPPENS IN CODE (DEVELOPER POV)	IMPACT ON TRAVELER (USER POV)
PREF-05 HIGH	`itinerary.service.ts:987–993` & `osm_places.json`	When `matchedPool` has fewer than 9 items for a 3-day trip, lines 987–993 append items from `curatedPool`. Even if sorted by preference, `osm_places.json` lists raw mountain peaks alphabetically (*Mount Agsil, Mount Aja, Mount Anao, Mount Anunas, Mount Balukbok, Mount Bonaba*), and there is no per-day meal slot or max-1-hike-per-day pacing rule.	Selecting **Nature & Parks** schedules **3 mountain hikes in a single day on Day 2 and 3 more on Day 3** with zero restaurants or lunch breaks.
PREF-06 HIGH	`itinerary.service.ts:996–998` & `itinerary_generator.py:376–382`	When `candidates.length < 3`, line 997 recycles already-used places from `matchedPool` ignoring `usedPlaceIds`. In addition, deduplication only checks `place.id` rather than normalized venue name or coordinate distance (`< 200m`).	In Test A, *Pantabangan Dam and Lake Viewpoint* is scheduled on Day 2 Stop 2 **and again on Day 3 Stop 1**, while *George Point Viewdeck & Resto* (Day 1) and *George Point* (Day 3) duplicate the same venue.
CUST-01 HIGH	`custom-builder.page.ts:551–564, 928–960`	In `proceedToBuilder()`, selecting `culture` or `resort` in Step 1 sets `this.selectedCategory = 'Tourist Spots'`. Yet in `applyBrowseFilter()` (line 930), `'Tourist Spots'` only allows `['tourist_attraction', 'nature_park', 'attraction']` —excluding `museum_cultural`, `church_religious`, and `resort_swimming`!	When a user selects **Culture & History** or **Resort** in Custom Itinerary Step 1, Step 2 opens on a filter tab that **hides all museums, churches, and resorts**!
CUST-02 HIGH	`custom-builder.page.ts:865–879`	`loadBrowsePlaces()` calls `placeService.getFeaturedPlaces(50)` (`featured: true`), loading only the **17 curated places** instead of querying the user's chosen Step 1 destination across the 2,673-POI database (`searchPlaces({ municipality_name: dest, limit: 100 })`).	In Custom Builder Step 2, users only see 17 places total (e.g., only 4 in Pantabangan, 0 in Gabaldon/Muñoz/Talavera) unless they manually type a search query.
CUST-03 MEDIUM	`custom-builder.page.ts:1165–1195`	In Step 3 Timeline, clicking `Auto-Schedule` chains stop times back-to-back (`end_time_of_stop_1 == start_time_of_stop_2`) without adding the inter-stop driving/tricycle travel duration between stops.	Custom itinerary timelines leave 0 minutes for road transit between destinations that may be 10–25 km apart.

4. User POV Improvement Proposal: UI & UX Design

USER EXPERIENCE REDESIGN

4.1 Categorized Destination Autofill Combobox & Regional Explorer Sheet

- **Instant Grouped Autofill Dropdown on Focus & Typing:** As soon as the user taps or types into the **Destination** field in either Suggested Itinerary or Custom Itinerary, open a clean floating combobox dropdown grouped into three scannable sections:
  1. **Province-Wide Option:** `🌏 All of Nueva Ecija (Best Multi-Town Corridor) — 2,673 places`
  2. **Municipalities & Cities (All 32 Nueva Ecija LGUs with POI Counts & Top Vibe Tags):** e.g., `📍 Pantabangan (Lake, Dam Viewpoints & Highland Resorts • 48 places)`, `📍 Cabanatuan City (Food Trip, Heritage & Cafes • 612 places)`, `📍 Gabaldon (Eco-Parks, Falls & River Resorts • 64 places)`, `📍 Science City of Muñoz (Agri-Tourism & Science Museums • 185 places)`, `📍 Gapan City (Lumang Gapan Heritage & Dining • 210 places)`, `📍 Bongabon (Farms, Cafes & Museums • 95 places)`.
  3. **Specific Landmarks & Attractions (Direct Anchor Selection):** e.g., `⭐ Minalungao National Park (General Tinio)`, `⭐ Highland Bali Villas Resort and Spa (Pantabangan)` —selecting a landmark automatically sets its municipality as the geographic cluster and locks that landmark as **Day 1 Stop 1** (`anchor_place_id`).
- **Replace Clipped Single-Row Chips with a Wrapped 2-Row Grid + `+ All 32 Towns` Trigger:** Replace the clipped `.quick-dest-row` (`overflow-x: auto`) with a wrapping pill grid (`flex-wrap: wrap`) showing the top 6 travel hubs in full text plus a `🗺️ Browse All 32 Towns` chip that opens a categorized North / Central / South / East Nueva Ecija picker modal.

4.2 Preference-Driven Pacing & Transparent Match Feedback in Suggested Wizard

- **Strict Preference Priority with Smart Meal Pacing:**
  - When a user selects specific preferences (e.g., `Nature & Parks` ONLY), every activity slot (`09:00 AM` morning and `02:30 PM` afternoon) must strictly pull from `Nature & Parks` (`nature_park` , `tourist_attraction`) within a dynamically expanding radius (`15 km` → `30 km` → `45 km`) before ever falling back to non-matching categories.
  - **Smart Dining Toggle ("Include Daily Lunch Spot"):** Add a lightweight toggle pill under Preferences (default: `ON`) so even when a traveler picks only `Nature & Parks` , each day still schedules 1 realistic midday dining stop (`12:00 PM – 01:30 PM`) near the morning attraction instead of scheduling 3 consecutive mountain peaks with no food!
  - **Activity Intensity Guard (Max 1 Mountain/Trail Hike Per Day):** Tag mountain peaks (`Mount Agsil` , `Mount Aja` , etc.) as `high_exertion` so the generator schedules at most 1 mountain/trail per day alongside scenic parks, viewpoints, or eco-farms.
- **Per-Stop "Why This Was Chosen" Preference Match Badge:** On every generated stop card in `/suggested-wizard` , render a subtle match pill—such as `✓ Matches: Resort • 2.1 km from previous stop` or `🍽️ Midday Lunch Anchor` — so the user immediately sees how their selected preferences shaped the itinerary.

4.3 Custom Itinerary Builder Step 1 → Step 2 Continuity

- **Multi-Preference Filter Sync & Destination Pre-Loading:** When the user clicks **Start Building Itinerary** → in Custom Builder Step 1, immediately query `GET /api/v1/places?municipality_name=<destination>&limit=100` from the live server (with 2,673-POI local OSM fallback) and add a `✨ Recommended for You` filter chip in Step 2 that combines *all* preferences chosen in Step 1 (`food` + `culture` + `resort`) rather than forcing a single `'Tourist Spots'` tab that hides resorts and museums.

5. Developer POV Improvement Proposal: Code & Algorithmic Blueprints ENGINEERING IMPLEMENTATION

5.1 Reusable `<app-destination-autocomplete>` Component (32 LGUs + Landmarks + Diacritic Normalization)

Create a standalone combobox component used by both `SuggestedWizardPage` and `CustomBuilderPage` that indexes all 32 Nueva Ecija municipalities/cities plus loaded landmarks using Unicode NFD diacritic stripping (so `"munoz"` matches `"Science City of Muñoz"` and `"penaranda"` matches `"Peñaranda"`):

```
// Project/src/app/shared/components/destination-autocomplete/destination-autocomplete.component.ts
export interface DestinationSuggestion {
  label: string;
  municipality: string;
  type: 'province' | 'municipality' | 'landmark';
  placeId?: string;
  subtitle: string;
  poiCount?: number;
}

export const NUEVA_ECIJA_32_LGUS: Array<{ name: string; district: string; highlights: string }> = [
  { name: 'Cabanatuan City', district: 'Central', highlights: 'Food Trip, Cafes & WWII Heritage' },
  { name: 'Pantabangan', district: 'North-East', highlights: 'Pantabangan Lake, Dam Viewpoints & Highland Resorts' }
  { name: 'Gabaldon', district: 'East', highlights: 'Dupinga River, Eco-Parks & Mountain Resorts' },
  { name: 'Bongabon', district: 'East', highlights: 'Farm Resorts, Rohi Cafe & Museo de Bongabon' },
  { name: 'Gapan City', district: 'South', highlights: 'Lumang Gapan Heritage Street & Local Dining' },
  { name: 'General Tinio', district: 'South-East', highlights: 'Minalungao National Park & Limestone River' },
  { name: 'San Jose City', district: 'North', highlights: 'Northern Gateway, Parks & Culinary Stops' },
  { name: 'Science City of Muñoz', district: 'North-West', highlights: 'PhilRice, CLSU Agri-Tourism & Museums' },
  { name: 'Palayan City', district: 'Central', highlights: 'Provincial Capitol Hub, Nature & Resorts' },
  { name: 'Talavera', district: 'Central', highlights: 'Highway Dining, Farm Parks & Resorts' },
  { name: 'Laur', district: 'East', highlights: 'Scenic Mountain Views & Nature Trails' },
  { name: 'Santa Rosa', district: 'Central-South', highlights: 'Family Resorts, Dining & Heritage Churches' },
  // ... + remaining 20 Nueva Ecija municipalities (Aliaga, Cabiao, Carranglan, Cuyapo, Guimba, Jaen, Licab,
  // Llanera, Lupao, Nampicuan, Peñaranda, Quezon, Rizal, San Antonio, San Isidro, San Leonardo, Santo Domingo, Talu
];

public normalizeText(str: string): string {
  return (str || '').normalize('NFD').replace(/[\u0300-\u036f]/g, '').toLowerCase().trim();
}
```

5.2 Fix Preference Category Mapping, Sequential Time Slots & Deduplication (`itinerary.service.ts` & `itinerary_generator.py`)

Replace the broken substring check (`a.categoryKey.toLowerCase().includes(pr)`) and hardcoded `time: '10:00 AM - 12:00 PM'` in `Project/src/app/core/services/itinerary.service.ts` with canonical category expansion, strict preference-

first filtering, fuzzy venue deduplication, and sequential travel-aware time slots:

```
// Project/src/app/core/services/itinerary.service.ts — Canonical Preference Map & Sequential Slot Scheduler
private readonly PREFERENCE_CATEGORY_MAP: Record<string, string[]> = {
  nature: ['nature_park', 'tourist_attraction'],
  food: ['restaurant', 'cafe'],
  culture: ['museum_cultural', 'church_religious'],
  shopping: ['shopping_market', 'convenience_store'],
  resort: ['resort_swimming', 'hotel_accommodation'],
};

private normalizeVenueName(name: string): string {
  return (name || '')
    .toLowerCase()
    .replace(/\b(viewdeck|viewpoint|and|&|resto|restaurant|cafe|resort|spa|hotel|villas)\b/g, '')
    .replace(/[^a-z0-9]/g, '');
}

private isMountainPeak(place: OfflineCuratedPlace): boolean {
  return /^(\bmt\.?\b|\bmount\b)/i.test(place.name.trim());
}

// Inside generateOfflineSuggestedItinerary():
// 1. Expand selected preferences into canonical category keys
const allowedCategories = new Set<string>();
for (const pr of prefs) {
  (this.PREFERENCE_CATEGORY_MAP[pr] || [pr]).forEach((c) => allowedCategories.add(c));
}

// 2. Assign sequential, non-overlapping time windows AFTER chronological/2-Opt ordering:
const SLOT_START_MINUTES = [9 * 60 + 30, 12 * 60 + 0, 15 * 60 + 0]; // 09:30 AM, 12:00 PM, 03:00 PM
let rollingMinutes = 9 * 60 + 30;
items.forEach((item, idx) => {
  item.sort_order = idx + 1;
  const dur = item.duration_minutes || 120;
  const travelMins = idx > 0 && item.travel_duration_seconds
    ? Math.max(15, Math.round(item.travel_duration_seconds / 60))
    : 0;
  const startMins = Math.max(SLOT_START_MINUTES[idx] ?? rollingMinutes, rollingMinutes + travelMins);
  const endMins = startMins + dur;
  item.planned_time = `${this.formatMinutesTo12Hr(startMins)} - ${this.formatMinutesTo12Hr(endMins)}`;
  rollingMinutes = endMins;
});
```

5.3 Decouple Synchronous Google API Calls in `itinerary_generator.py` & Fix Custom Builder Step 2 Filter

- **Backend Fast-Path Generation (`backend/app/algorithms/itinerary_generator.py:321, 507`):** Default `include_live_places` to `False` when the local PostgreSQL/SQLite database already has `>= 15` candidates within the 25 km bounding box, and compute Haversine/OSRM polylines locally unless explicit Google Routes enrichment is requested. Increase `itinerary.service.ts:360` timeout from `timeout(4000)` to `timeout(10000)` so 3-day server generations never abort prematurely into offline mode.
- **Fix Custom Builder Step 1 → Step 2 Filter & Initial Load (`custom-builder.page.ts:551–565, 865–879`):** In `loadBrowsePlaces()`, call `this.placeService.searchPlaces({ municipality_name: dest !== 'Nueva Ecija' ? dest : undefined, limit: 100 })` instead of `getFeaturedPlaces(50)`, and map `culture` to `'Culture & Heritage'` and `resort` to `'Resorts & Stays'` (or keep `selectedCategory = 'All'` sorted by matching Step 1 preference categories at the top).

6. Phased Implementation Roadmap & Verification Matrix

DELIVERY PLAN

PHASE	DELIVERABLE & SCOPE	TARGET FILES	VERIFICATION GATE
Phase 1 IMMEDIATE	**Destination Autofill Combobox & Wrapped Quick-Select Pills**	`destination-autocomplete.component.ts`, `suggested-wizard.page.html`, `custom-builder.page.html`	Typing `"panta"`, `"munoz"`, or `"minalungao"` opens grouped municipality + landmark suggestions; 0 clipped chips on mobile/desktop.

PHASE	DELIVERABLE & SCOPE	TARGET FILES	VERIFICATION GATE
Phase 2 IMMEDIATE	**Fix Preference Category Map, Sequential Time Slots & Deduplication**	`itinerary.service.ts`, `backend/app/algorithms/itinerary_generator.py`	Selecting `Nature & Parks` changes Day 1 Stop 1 immediately, schedules max 1 mountain peak/day + 1 lunch stop, produces 0 duplicate venues across days, and assigns sequential non-overlapping times (`09:30 AM` , `12:00 PM` , `03:00 PM`).
Phase 3 SHORT-TERM	**Custom Builder Full-Catalog Browse & Preference Filter Fix**	`custom-builder.page.ts` (`proceedToBuilder` , `loadBrowsePlaces` , `applyBrowseFilter`)	Selecting `Culture & History` or `Resort` in Step 1 surfaces matching museums, churches, and resorts in Step 2 across all 2,673 POIs rather than only 17 featured places.